THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 11: **Patterns in Life: Diversity and Classification** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An organism is unicellular, eukaryotic and has chloroplasts and a flagellum. It belongs to: 1

(a) Fungi **(b)** Protista **(c)** Monera **(d)** Plantae

2. A fish differs from an amphibian in having: 1

(a) a three-chambered heart **(b)** gills throughout life

(c) moist skin **(d)** lungs

3. The correct sequence of taxonomic ranks for the tiger from the highest is: 1

(a) Panthera, Felidae, Carnivora, Mammalia

(b) Animalia, Mammalia, Chordata, Felidae, Carnivora, Panthera

(c) Chordata, Animalia, Mammalia, Carnivora

(d) Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris

4. A plant is described as having vascular bundles scattered in the stem, parallel venation and fibrous roots. It is a: 1

(a) monocot **(b)** gymnosperm **(c)** dicot **(d)** pteridophyte

5. An organism is multicellular, has a cell wall of chitin and feeds on dead matter. It is a: 1

(a) plant **(b)** fungus **(c)** bacterium **(d)** animal

6. An animal that lays eggs, has feathers and a beak is a: 1

(a) amphibian **(b)** mammal **(c)** bird **(d)** reptile

7. Two species placed in the same genus are: 1

(a) of different kingdoms

(b) more closely related than species in the same family but different genera

(c) unrelated

(d) identical

8. The seed of wheat is a: 1

(a) dicot seed **(b)** gymnosperm seed **(c)** monocot seed **(d)** spore

9. An organism has a notochord, a dorsal nerve cord and gill slits at some stage. It is a: 1

(a) arthropod **(b)** mollusc **(c)** chordate **(d)** annelid

10. A mammal that lays eggs is the: 1

(a) platypus **(b)** bat **(c)** kangaroo **(d)** dolphin

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A student has a sample of a unicellular organism with a nucleus and cilia. In which kingdom will you place it and why? 2

12. Why is a whale classified with mammals though it lives in water? 2

13. How do you distinguish a roundworm from an earthworm? 2

14. Why are viruses not included in the five-kingdom classification? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Classify the following into the correct groups: banyan, fern, Spirogyra, moss, pine, rice. 3

16. An animal has a four-chambered heart, feathers and hollow bones. Identify the class and mention how the bones are adapted. 3

17. Describe the features of Platyhelminthes and the diseases caused by the parasitic forms. 3

OR

Classify the following animals into phyla: earthworm, hydra, octopus, tapeworm, starfish, cockroach. 3

18. Explain the concept of a species and why interbreeding is important to define it. 3

OR

Explain why the evolutionary classification arranges organisms from simple to complex with the example of animals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student collects ten organisms: mushroom, rose, whale, bacterium, Euglena, fern, snake, pigeon, earthworm and moss. Classify each into the kingdom and the group. 5

OR

Describe the adaptations of the birds for flight and of the fish for aquatic life with examples. 5

20. Explain the characteristics of the five classes of vertebrates in a table of five characters: skin, respiration, heart, reproduction, body temperature. 5

OR

Explain the importance of biodiversity to humans and the ecosystem and the measures to conserve it. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND